

How the symmetry-adapted blocks are related, and how much unique information they carry

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What this note is about

This summarizes how the pieces of the space-group change-of-basis Q (the transform taking the ISDF grid index (k, I) to symmetry-adapted labels (i, a, c)) are related to one another, and how little *genuinely independent* information Q contains. The running example is diamond ($n_{\text{IP}} = 136$ interpolation points per cell, 48 point-group operations), on $2 \times 2 \times 2$ and $6 \times 6 \times 6$ k -meshes. Everything below is nonsymmorphic-safe: the lattice-vector / fractional-translation phases are carried explicitly, and at Brillouin-zone boundaries the little-group representations are the projective ones.

1 Block structure of Q

Q is a square unitary of size $N = n_k n_{\text{IP}}$. Two structural facts:

1. **Block-diagonal in k .** Each symmetry-adapted vector has a definite momentum, so $Q = \bigoplus_k Q^{(k)}$ with n_k blocks, each $n_{\text{IP}} \times n_{\text{IP}}$. Q never mixes different momenta.
2. **Grouped into families (stars).** The point-group operations permute the k -points into *families*; within a family the members are carried into one another by “move-it” operations. Only **one block per family** is independent.

Within a representative block $Q^{(k_0)}$, the transform is a direct sum of small little-group “pattern” blocks, one per interpolation-point orbit (size $\leq |L_k| \leq 48$).

2 How the blocks of a family are related

Fix a family with representative momentum k_0 and another member $k_j = g_j \cdot k_0$, where g_j is a chosen move-it operation. For any symmetry-adapted column c , the entries of the two blocks are related **entry-by-entry** by

$$Q_{\text{perm}[g_j, I], c}^{(k_j)} = e^{-i k_j \cdot L_I(g_j)} Q_{I, c}^{(k_0)}$$

i.e. as matrices $Q^{(k_j)} = P(g_j) \Lambda(g_j, k_j) Q^{(k_0)}$, where $P(g_j)$ permutes the interpolation-point rows by $\text{perm}[g_j, \cdot]$ and Λ is the diagonal Bloch phase $e^{-i k_j \cdot L_I(g_j)}$. Reading it off, across a family and for the same column c :

- **rows are permuted:** $I \mapsto \text{perm}[g_j, I]$ (the grid shuffle of g_j);
- **each entry gets a phase** $e^{-i k_j \cdot L_I(g_j)}$ — the only k -dependence;
- **magnitudes are unchanged:** $|Q_{\text{perm}[g_j, I], c}^{(k_j)}| = |Q_{I, c}^{(k_0)}|$;
- **the pattern label c is preserved** (same irrep, same little-group partner, same copy); only the star-member advances $k_0 \rightarrow k_j$.

Here $L_I(g)$ is the **fold-back lattice vector**, defined by $g \cdot r_I = r_{\text{perm}[g,I]} + L_I(g)$ (for nonsymmorphic $g = \{R \mid \tau\}$ the fractional translation τ is included in $g \cdot r_I$). It depends only on (g, I) , *not* on k ; the k -dependence is entirely the computable phase $e^{-ik \cdot L}$. Different families are independent (each keeps its own representative block).

3 Amount of unique information

Because every non-representative block is a phased-permutation *image* of its family representative, the genuinely independent content of Q is just the representative blocks:

$$\#\{\text{unique numbers}\} = \sum_{\text{family reps}} \sum_{\text{orbits } o} |o|^2.$$

The auxiliary objects g_j (which operation reaches which member) and $L_I(g)$ are **not** per-family data: they are properties of the 48 operations and the grid geometry, shared across all families and all k . In fact they follow from a tiny, k -independent **generation kernel**:

$$\underbrace{n_{\text{IP}} \times 3}_{\text{point positions}} + \underbrace{48 \times (9 + 3)}_{\text{ops } (R, \tau)} \approx 1000 \text{ numbers},$$

from which perm, L , all phases, and the k -map are computed. Per family one records *only* its representative block.

The numbers, and how they scale

	$2 \times 2 \times 2$	$6 \times 6 \times 6$	growth
n_k (momenta)	8	216	$27 \times$
families (stars)	3	16	$5.3 \times$
unique data of Q ($\sum o ^2$)	6,800	12,912	$1.9 \times$
block-diagonal-in- k $n_k n_{\text{IP}}^2$	147,968	3,995,136	$27 \times (\propto n_k)$
dense $(n_k n_{\text{IP}})^2$	1,183,744	862,949,376	$729 \times (\propto n_k^2)$

Key point. From $2 \times 2 \times 2$ to $6 \times 6 \times 6$ the number of momenta grows $27 \times$, but the genuine unique data grows only $\approx 1.9 \times$. It tracks the number of *families*, not n_k — and even that sublinearly, because the many new k -points are low-symmetry (little groups of size 2 or 4), whose small little groups split the 136 points into many tiny orbits that contribute almost nothing to $\sum |o|^2$ (the bulk is still the few high-symmetry families, e.g. Γ alone gives 3872). For $6 \times 6 \times 6$ the unique data (12,912) is about $67,000 \times$ smaller than the dense matrix (8.6×10^8).

Summary

- Q is block-diagonal in k ; its blocks come in **families**, and only one block per family is independent.
- Within a family, member blocks are **phased-permutation images** of the representative: same magnitudes, rows permuted by g_j , times a member-specific Bloch phase $e^{-ik_j \cdot L}$; the pattern label is preserved.

- The unique information is $\sum_{\text{families}} \sum_{\text{orbits}} |o|^2$ (6,800 for $2 \times 2 \times 2$, 12,912 for $6 \times 6 \times 6$), plus a ~ 1000 -number, k -independent generation kernel. The k -dependence lives only in computable phases, so the information is **essentially flat in n_k** — it scales with the number of families, not the number of k -points.